

Ellen Poulsen

 Ellen Poulsen |  ecpoulse@calpoly.edu |  +1 (541) 654-2129

EDUCATION

2025 — Present **B.S. Aerospace Engineering** (GPA: 3.66/4.0)
California Polytechnic State University, San Luis Obispo

2023 Recipient of the Oregon State Seal of Biliteracy in Spanish

Relevant Coursework: Introduction to Design and Manufacturing, Welding, Statics & Mechanics, Stars & Planetary Systems, Linear Analysis (expected Spring 2027), Dynamics (expected Spring 2027)

WORK EXPERIENCE

Test Engineering Intern — Empirical Systems Aerospace *June 2026 — Aug. 2026*
San Luis Obispo, CA

- Developed servo testing procedures and software to analyze frequency response for integration on flight vehicles.
- Programmed servo control mechanisms utilizing both PWM and CAN bus communication protocols.
- Calibrated data acquisition timing across multiple testing devices to ensure precise sensor telemetry collection.
- Managed project version control and code collaboration using Python, VS Code, and GitLab.

Observatory Assistant — Cal Poly Physics Department *May 2026 — Present*

- Operating 14-inch Meade LX600 ACF Telescope for public viewing events.
- Led educational observing sessions for student groups, presenting key astrophysics concepts and managing night-sky targets.

PROJECTS

Cal Poly Space Systems — Liquids Project

- Development of bipropellant liquid rocket engines, test stand, and associated test procedures and safety
- Familiarization with Nitrous Oxide and Ethanol as propellants and work with high pressure systems
- Static pressure testing, hot fires, cold flows, and work on various projects including regenerative rocket engine and test stand maintenance

LEADERSHIP EXPERIENCE

Cal Poly Space Systems Lead — SILLE (Student Introductory Liquids Learning Engine)

- Guiding students through project to design and manufacture their own small liquid rocket engine to test up to 50lbs thrust
- Organization of over 40 students into teams, teaching basics of liquids rocketry, foundational understanding of isentropic equations and associated variables, machining, and safe testing practices
- Public speaking and proficiency in communication utilizing Microsoft teams, powerpoint, forms, and excel
- Mentorship with focus on member retention and transfer of knowledge

Society of Women Engineers at Cal Poly

- Advocating for and supporting women underrepresented in fields of engineering.
- Community outreach and educational opportunities to advance access to STEM and inspire the next generation of girls in engineering